Supplement: all arms, folds and cohort domains

Supplementary material, 28 October 2026

The six tables retain all 90 combinations of five arms, three outer folds, two cohort domains and three input conditions. They use the frozen 75-fit execution and independently audited saved probabilities. E1–E3 identify the excluded environment, including when describing familiar-test participants from the remaining two environments. Each held fold has 420 recordings; each familiar fold has 168. The participant roles are disjoint within a fold.

Every entry is a mean and sample standard deviation across five training seeds. Five masks are averaged within a seed before this summary under each loss condition; complete input has one realization. These are not population confidence intervals. Familiar cohorts may recur across outer folds, and repeated masks are dependent views of fixed recordings. The extension is exploratory following historically known target outcomes. All unfavourable results are retained.

The immutable numeric source is `per_fold_domain_mean_sampleSD_5seeds.csv`; its full SHA256 is recorded with the distributed result tables and checksums. NLL and Brier use the same source-validation temperature across all test conditions. Brier is summed over seven classes, not divided by seven.

Table S1: All arms and outer folds, held-environment cohort, complete input. Each entry is mean ± sample SD across five training seeds; mask means are formed first. Each held fold contains 420 recordings from 10 excluded-environment participants. E1–E3 identify the excluded environment, including for the familiar cohort. Accuracy and macro F1 are percentages.

Arm	Fold	Accuracy (%)	Macro F1 (%)	NLL	Brier
A	E1	52.19 ± 5.30	50.01 ± 5.73	1.378 ± 0.078	0.621 ± 0.024
A	E2	45.10 ± 2.48	43.66 ± 4.35	1.464 ± 0.136	0.760 ± 0.048
A	E3	47.62 ± 4.46	46.69 ± 4.86	1.345 ± 0.146	0.689 ± 0.063
B	E1	51.71 ± 4.29	49.31 ± 4.59	1.399 ± 0.094	0.627 ± 0.020
B	E2	45.52 ± 2.49	43.88 ± 4.57	1.475 ± 0.126	0.765 ± 0.049
B	E3	46.90 ± 4.50	45.88 ± 5.04	1.362 ± 0.149	0.695 ± 0.063
C	E1	49.81 ± 3.58	47.29 ± 3.61	1.538 ± 0.111	0.672 ± 0.027
C	E2	40.81 ± 3.19	40.06 ± 3.57	1.507 ± 0.155	0.797 ± 0.053
C	E3	43.00 ± 2.65	42.24 ± 2.40	1.528 ± 0.121	0.747 ± 0.046
D	E1	28.33 ± 3.95	24.80 ± 3.89	2.044 ± 0.050	0.865 ± 0.017
D	E2	21.24 ± 0.78	19.68 ± 1.66	2.160 ± 0.129	0.950 ± 0.039
D	E3	26.10 ± 1.29	23.77 ± 1.37	2.022 ± 0.087	0.877 ± 0.025
E	E1	71.24 ± 1.20	69.50 ± 1.46	1.070 ± 0.066	0.450 ± 0.021
E	E2	75.38 ± 1.29	75.01 ± 1.51	0.714 ± 0.030	0.350 ± 0.009
E	E3	78.90 ± 1.95	78.74 ± 2.12	0.534 ± 0.030	0.294 ± 0.020

Table S2: All arms and outer folds, held-environment cohort, random 50% loss. Each entry is mean ± sample SD across five training seeds; mask means are formed first. Each held fold contains 420 recordings from 10 excluded-environment participants. E1–E3 identify the excluded environment, including for the familiar cohort. Accuracy and macro F1 are percentages.

Arm	Fold	Accuracy (%)	Macro F1 (%)	NLL	Brier
A	E1	54.58 ± 3.58	52.74 ± 3.66	1.330 ± 0.074	0.600 ± 0.021
A	E2	47.30 ± 2.44	46.87 ± 3.58	1.361 ± 0.111	0.716 ± 0.048
A	E3	46.85 ± 3.99	46.08 ± 4.63	1.290 ± 0.091	0.677 ± 0.045
B	E1	54.22 ± 2.95	51.86 ± 3.46	1.381 ± 0.134	0.606 ± 0.027
B	E2	47.16 ± 2.36	45.92 ± 3.90	1.400 ± 0.109	0.730 ± 0.055
B	E3	46.31 ± 3.86	45.68 ± 4.52	1.301 ± 0.099	0.683 ± 0.047
C	E1	49.51 ± 2.56	45.99 ± 2.54	1.586 ± 0.093	0.677 ± 0.021
C	E2	41.62 ± 2.96	41.25 ± 3.69	1.505 ± 0.138	0.776 ± 0.052
C	E3	39.84 ± 1.89	38.50 ± 2.36	1.512 ± 0.100	0.747 ± 0.042
D	E1	28.05 ± 2.54	24.30 ± 2.48	2.027 ± 0.063	0.863 ± 0.014
D	E2	19.16 ± 1.70	16.45 ± 2.08	2.084 ± 0.134	0.920 ± 0.038
D	E3	23.07 ± 3.83	19.53 ± 4.03	2.042 ± 0.105	0.883 ± 0.031
E	E1	69.78 ± 1.30	68.04 ± 1.50	1.149 ± 0.076	0.468 ± 0.022
E	E2	74.49 ± 1.16	73.98 ± 1.53	0.760 ± 0.046	0.368 ± 0.016
E	E3	77.20 ± 2.51	77.09 ± 2.71	0.584 ± 0.039	0.321 ± 0.024

Table S3: All arms and outer folds, held-environment cohort, contiguous 50% loss. Each entry is mean ± sample SD across five training seeds; mask means are formed first. Each held fold contains 420 recordings from 10 excluded-environment participants. E1–E3 identify the excluded environment, including for the familiar cohort. Accuracy and macro F1 are percentages.

Arm	Fold	Accuracy (%)	Macro F1 (%)	NLL	Brier
A	E1	39.70 ± 2.63	37.16 ± 3.26	1.978 ± 0.092	0.817 ± 0.020
A	E2	30.04 ± 0.95	29.31 ± 2.51	2.218 ± 0.190	0.987 ± 0.035
A	E3	34.59 ± 3.84	32.95 ± 4.12	2.184 ± 0.237	0.903 ± 0.058
B	E1	39.12 ± 2.64	36.28 ± 3.34	1.975 ± 0.084	0.815 ± 0.018
B	E2	29.99 ± 0.83	28.96 ± 2.41	2.244 ± 0.169	0.993 ± 0.028
B	E3	33.45 ± 4.14	32.22 ± 4.21	2.211 ± 0.249	0.915 ± 0.062
C	E1	36.71 ± 1.42	33.41 ± 1.92	2.197 ± 0.140	0.872 ± 0.025
C	E2	29.09 ± 1.42	29.18 ± 2.30	2.176 ± 0.217	0.979 ± 0.048
C	E3	31.13 ± 2.21	29.25 ± 2.30	2.243 ± 0.193	0.921 ± 0.054
D	E1	25.14 ± 2.78	21.52 ± 2.17	2.379 ± 0.157	0.926 ± 0.034
D	E2	17.35 ± 2.11	15.71 ± 1.80	2.711 ± 0.245	1.080 ± 0.059
D	E3	22.12 ± 2.14	19.21 ± 1.75	2.463 ± 0.134	0.985 ± 0.039
E	E1	60.49 ± 1.24	57.87 ± 2.10	1.285 ± 0.044	0.570 ± 0.013
E	E2	61.09 ± 1.41	59.51 ± 1.29	1.189 ± 0.045	0.552 ± 0.016
E	E3	60.22 ± 2.90	59.47 ± 2.85	1.007 ± 0.043	0.536 ± 0.025

Table S4: All arms and outer folds, familiar-environment cohort, complete input. Each entry is mean ± sample SD across five training seeds; mask means are formed first. Each familiar fold contains 168 recordings from four source-environment participants excluded from fitting and validation. E1–E3 identify the excluded environment, including for the familiar cohort. Accuracy and macro F1 are percentages.

Arm	Fold	Accuracy (%)	Macro F1 (%)	NLL	Brier
A	E1	55.71 ± 6.06	54.68 ± 5.85	1.093 ± 0.080	0.567 ± 0.046
A	E2	42.50 ± 5.37	41.09 ± 5.61	1.788 ± 0.262	0.802 ± 0.081
A	E3	51.07 ± 1.60	48.67 ± 1.92	1.326 ± 0.098	0.661 ± 0.025
B	E1	55.24 ± 4.93	54.19 ± 4.82	1.111 ± 0.076	0.577 ± 0.043
B	E2	42.74 ± 5.53	41.68 ± 5.95	1.795 ± 0.250	0.802 ± 0.080
B	E3	51.07 ± 1.14	48.98 ± 1.50	1.331 ± 0.097	0.665 ± 0.027
C	E1	57.74 ± 4.85	55.89 ± 4.48	1.156 ± 0.067	0.570 ± 0.044
C	E2	43.45 ± 8.10	42.59 ± 8.13	1.740 ± 0.312	0.786 ± 0.108
C	E3	51.19 ± 1.52	48.49 ± 1.82	1.370 ± 0.103	0.664 ± 0.030
D	E1	33.93 ± 3.09	32.06 ± 3.32	1.656 ± 0.038	0.772 ± 0.013
D	E2	26.67 ± 2.12	24.63 ± 2.19	2.022 ± 0.159	0.889 ± 0.046
D	E3	30.48 ± 3.63	28.24 ± 4.27	1.808 ± 0.086	0.833 ± 0.027
E	E1	75.83 ± 1.91	75.13 ± 2.30	0.650 ± 0.027	0.343 ± 0.011
E	E2	63.57 ± 2.43	62.68 ± 2.71	1.166 ± 0.117	0.549 ± 0.043
E	E3	65.83 ± 2.51	65.05 ± 2.92	0.832 ± 0.043	0.453 ± 0.019

Table S5: All arms and outer folds, familiar-environment cohort, random 50% loss. Each entry is mean ± sample SD across five training seeds; mask means are formed first. Each familiar fold contains 168 recordings from four source-environment participants excluded from fitting and validation. E1–E3 identify the excluded environment, including for the familiar cohort. Accuracy and macro F1 are percentages.

Arm	Fold	Accuracy (%)	Macro F1 (%)	NLL	Brier
A	E1	55.07 ± 5.44	54.17 ± 4.77	1.080 ± 0.071	0.561 ± 0.045
A	E2	41.14 ± 5.36	39.80 ± 5.40	1.710 ± 0.155	0.784 ± 0.059
A	E3	51.79 ± 1.42	50.09 ± 1.00	1.267 ± 0.078	0.636 ± 0.024
B	E1	54.14 ± 5.40	53.17 ± 4.69	1.094 ± 0.074	0.565 ± 0.042
B	E2	41.90 ± 4.71	40.84 ± 5.03	1.711 ± 0.139	0.782 ± 0.057
B	E3	50.90 ± 1.58	49.26 ± 0.99	1.286 ± 0.075	0.647 ± 0.025
C	E1	54.00 ± 4.47	51.42 ± 3.60	1.232 ± 0.067	0.594 ± 0.039
C	E2	39.69 ± 7.85	37.62 ± 7.81	1.804 ± 0.238	0.794 ± 0.076
C	E3	49.60 ± 1.56	47.28 ± 1.82	1.397 ± 0.100	0.670 ± 0.030
D	E1	32.33 ± 1.21	29.69 ± 1.97	1.693 ± 0.030	0.785 ± 0.009
D	E2	23.74 ± 4.40	20.24 ± 5.01	2.007 ± 0.105	0.876 ± 0.040
D	E3	27.12 ± 4.11	23.89 ± 3.59	1.816 ± 0.072	0.834 ± 0.021
E	E1	74.14 ± 1.83	73.56 ± 2.12	0.687 ± 0.027	0.358 ± 0.015
E	E2	58.71 ± 2.60	57.49 ± 2.79	1.297 ± 0.136	0.597 ± 0.039
E	E3	66.33 ± 1.31	65.52 ± 1.78	0.854 ± 0.049	0.456 ± 0.016

Table S6: All arms and outer folds, familiar-environment cohort, contiguous 50% loss. Each entry is mean ± sample SD across five training seeds; mask means are formed first. Each familiar fold contains 168 recordings from four source-environment participants excluded from fitting and validation. E1–E3 identify the excluded environment, including for the familiar cohort. Accuracy and macro F1 are percentages.

Arm	Fold	Accuracy (%)	Macro F1 (%)	NLL	Brier
A	E1	45.07 ± 3.92	44.29 ± 3.61	1.564 ± 0.060	0.738 ± 0.041
A	E2	33.60 ± 2.21	33.28 ± 2.28	2.284 ± 0.278	0.939 ± 0.048
A	E3	41.81 ± 1.35	39.22 ± 1.81	1.863 ± 0.082	0.791 ± 0.032
B	E1	44.19 ± 3.87	43.76 ± 3.48	1.565 ± 0.059	0.739 ± 0.040
B	E2	33.14 ± 2.44	32.85 ± 2.49	2.343 ± 0.252	0.953 ± 0.041
B	E3	41.86 ± 1.14	39.46 ± 1.90	1.881 ± 0.085	0.801 ± 0.032
C	E1	45.74 ± 3.26	43.17 ± 2.61	1.632 ± 0.087	0.749 ± 0.039
C	E2	31.86 ± 5.26	30.84 ± 5.69	2.354 ± 0.391	0.949 ± 0.088
C	E3	43.95 ± 3.11	41.32 ± 3.45	1.895 ± 0.119	0.788 ± 0.035
D	E1	29.36 ± 2.14	26.99 ± 1.74	1.835 ± 0.076	0.826 ± 0.021
D	E2	19.90 ± 3.35	17.14 ± 3.89	2.604 ± 0.254	1.018 ± 0.070
D	E3	25.48 ± 3.40	22.51 ± 2.67	2.203 ± 0.151	0.936 ± 0.027
E	E1	56.02 ± 0.66	52.76 ± 0.86	1.207 ± 0.058	0.604 ± 0.021
E	E2	48.52 ± 2.83	46.63 ± 3.64	1.593 ± 0.163	0.731 ± 0.048
E	E3	55.43 ± 2.22	54.14 ± 2.56	1.223 ± 0.044	0.609 ± 0.021